Introduction and Math Review Practice 2

Module 1 - Introduction & Math Review · Daily practice sheet · No date

Name _____ Date _____

Problem 1 How many half-lives ($t_{1/2}$) would it take for 99.9% of any initial concentration of a drug to decompose? Assume first-order kinetics.

Problem 2 If the half-life for decomposition of a drug is 12 hours, how long will it take for 125 mg of the drug to decompose by 30%? Assume first-order kinetics and constant temperature.

Problem 3 Exactly 300 mg of a drug are dissolved into an unknown volume of distilled water. After complete dissolution of the drug, 1.0-mL samples were removed and assayed for the drug. The following results were obtained:

Time (hr)	Concentration (mg/mL)
0.5	0.45
2.0	0.3

Assuming zero-order decomposition of the drug, what was the original volume of water in which the drug was dissolved?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Problem 1 Her key

Answer: 10 half-lives

WHAT THE STEM GIVES

fraction	99.9%	decomposed, so 0.1% remains: $C_0/C = 1000$

THE WORKING, ONE LINE AT A TIME

SETUP

99.9% decomposed leaves 0.1% remaining: $C_0 = 100$, $C = 0.1$

The first-order equation works with what remains, so the 99.9% decomposed becomes the 0.1% left. Any starting value gives the same count; 100 lets the percentages be read directly.

SETUP

$$t = \frac{\ln(C_0 \div C)}{k}, \text{ with } k = \frac{0.693}{t½}$$

$C = C_0\, e^{-kt}$ solved for t gives $\ln(\frac{C_0}{C})$ over k. Putting $\frac{0.693}{t½}$ in place of k writes the time as a multiple of t½, which is what the question asks for.

ALGEBRA

$$\ln(\frac{100}{0.1}) = \ln(1000) = 6.908;\ \frac{6.908}{0.693} = 9.97, \text{ so } t = 9.97\ t½$$

C_0 over C is a ratio of two concentrations, so the units cancel and its logarithm is a pure number; dividing by 0.693 counts the half-lives.

ROUND

t = 10 half-lives

Her key takes 9.97 as 10 and confirms it with her table of halvings: after 10 half-lives 0.1% remains, so 99.9% has decomposed, whatever the starting concentration.

> **DOES THE NUMBER MAKE SENSE?**
> Ten halvings leave about one thousandth, 0.1%, so 99.9% has decomposed after about ten half-lives; 9.97 rounds to 10.

Her key gives 9.97 t½ from the equation and 10 half-lives from her table; both are accepted.

Problem 2 Her key

Answer: 6.2 hr

WHAT THE STEM GIVES

t½	12 hours	gives $k = \frac{0.693}{12} = 0.05775\ hr^{-1}$
A0	125 mg	the starting amount, top of the log ratio
decomposed	30%	leaves 70%: A = 87.5 mg
temperature	constant	*not needed: it keeps k fixed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{0.693}{12\ \text{hr}} = 0.05775\ \text{hr}^{-1}$$

$k = \frac{0.693}{t_{1/2}}$: 0.693 is a pure number, so dividing by hours leaves reciprocal hours, the unit of a first-order rate constant.

ALGEBRA

30% of 125 mg decomposed leaves (0.7)(125 mg) = 87.5 mg

The equation works with the amount remaining: 30% gone leaves 70% of the 125 mg.

SETUP

$$A = A0\ e^{-kt}, \text{ so } t = \frac{\ln(A0 \div A)}{k}$$

Taking the natural log of the first-order equation and solving for t gives the time for the amount to fall from A0 to A.

ALGEBRA

$$t = \frac{\ln(125 \div 87.5)}{0.05775\ \text{hr}^{-1}} = \frac{0.3567}{0.05775\ \text{hr}^{-1}} = 6.18\ \text{hr}$$

The milligrams cancel inside the logarithm, leaving a pure number, 0.3567; dividing it by reciprocal hours leaves hours.

ROUND

t = 6.2 hr

Her key reports 6.2 hr, to one decimal place. Less than one 12-hour half-life has passed, as it must when less than half of the drug has decomposed.

> **DOES THE NUMBER MAKE SENSE?**
> Less than half has decomposed, so the time is less than one half-life, 12 hours; 30% is a little over half of the 50% one half-life removes, so about 6 hours fits.

Her key rounds k to 0.058 hr^{-1}. Dividing by 0.058 gives 6.15 hr; the unrounded k, 0.05775 hr^{-1}, gives 6.18 hr, her 6.2 hr. Both are accepted.

Problem 3 Her key

Answer: 600 mL

WHAT THE STEM GIVES

dose	300 mg	the numerator of $V = \frac{dose}{C_0}$
sample	1.0-mL	*not needed: the size of each sample*
C at 0.5 hr	0.45 mg/mL	the earlier point of the slope
C at 2.0 hr	0.3 mg/mL	the later point; $C_0 = 0.3 + k \times 2$

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = -\frac{\Delta Y}{\Delta X} = -\frac{(0.3 - 0.45)\ \text{mg/mL}}{(2 - 0.5)\ \text{hr}} = 0.1\ \text{mg/mL/hr}$$

Zero order is a straight line, $C = C_0 - kt$, so k is the slope with the sign reversed. mg/mL over hours gives mg/mL/hr: a fixed concentration lost each hour.

ALGEBRA

$$A0 = 0.3\ \text{mg/mL} + (0.1\ \text{mg/mL/hr})(2\ \text{hr}) = 0.5\ \text{mg/mL}$$

$C_0 = C + kt$ adds back what was lost in the 2 hours before the second sample. The hours cancel in (mg/mL/hr)(hr), so both terms are in mg/mL and add.

SETUP

$$\frac{300\ \text{mg}}{X\ \text{mL}} = 0.5\ \text{mg/mL}$$

At time zero all 300 mg was in solution, so concentration = amount over volume, the line $C_p = \frac{D_B}{V_D}$, fixes the unknown volume X.

ALGEBRA

$$X = \frac{300\ \text{mg}}{0.5\ \text{mg/mL}} = 600\ \text{mL}$$

mg divided by mg/mL leaves mL. The volume is larger than 300 mL because the starting concentration is below 1 mg/mL.

DOES THE NUMBER MAKE SENSE?

Losing 0.1 mg/mL each hour, the solution held 0.5 mg/mL at time zero, so 300 mg sits in 600 mL. C_0 is above the 0.45 mg/mL measured at 0.5 hours, so the volume is below 300 mg ÷ 0.45 mg/mL.